

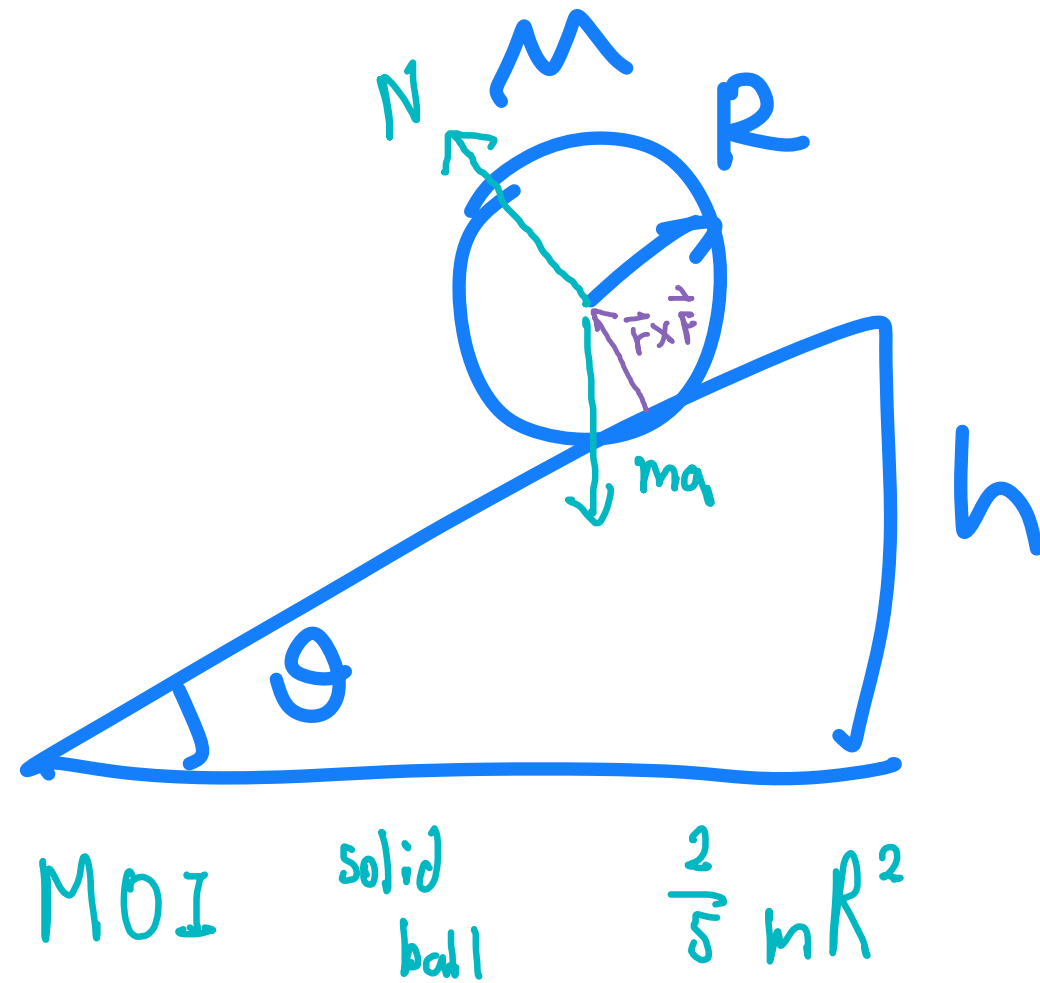


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PHY1001 Mechanics

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Acceleration of A Rolling Ball



A solid ball of mass M and radius R rolls down a ramp without slipping. The ramp has an angle θ w.r.t. horizontal plane. The surface is not smooth. Find the velocity of center of mass when the solid ball reaches the end of the ramp. $\rightarrow$ friction

Newton
2nd law
(rotation) $\tau = mg \cdot r \cdot \sin\theta = I_p \alpha$ (the only force here: gravity)

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$|\vec{\tau}| = |\vec{r}| |\vec{F}| \sin\theta$$

parallel axis theorem

$$I_p = I_{cm} + MR^2$$

$$= \frac{7}{5} MR^2$$

roll without slipping $R\alpha = a$

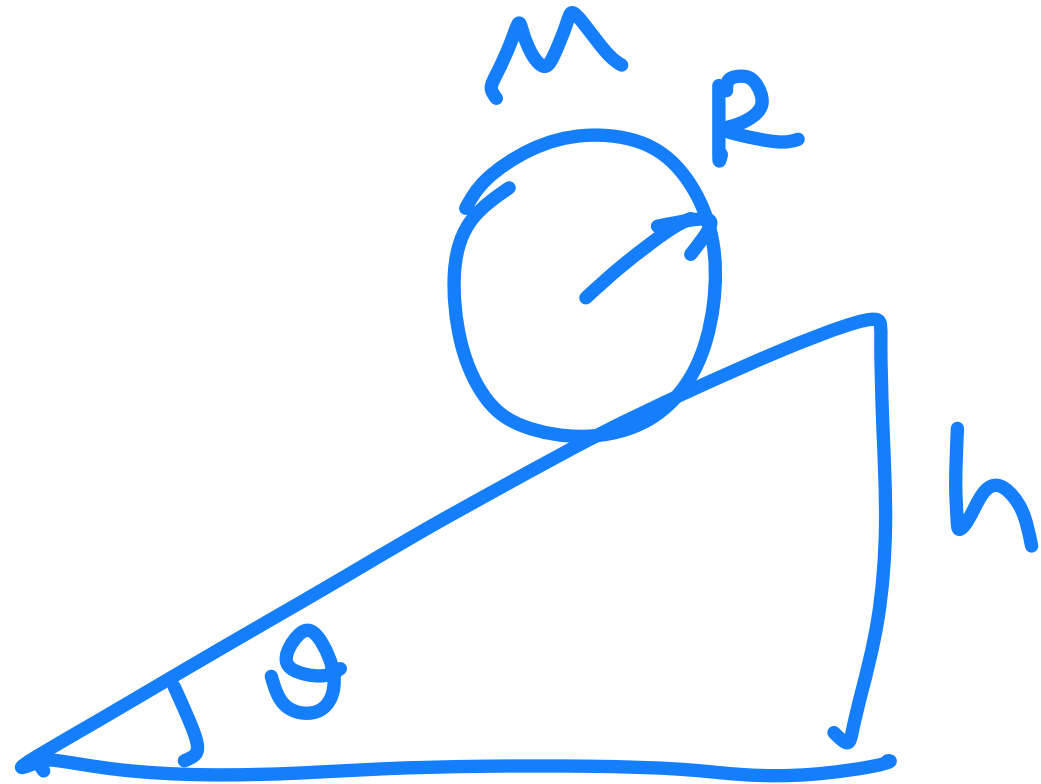
$$v^2 = 2as$$

$$v = \sqrt{2a \frac{h}{\sin\theta}}$$

$$v = \sqrt{\frac{10gh}{7}}$$

1D motion with
constant acceleration

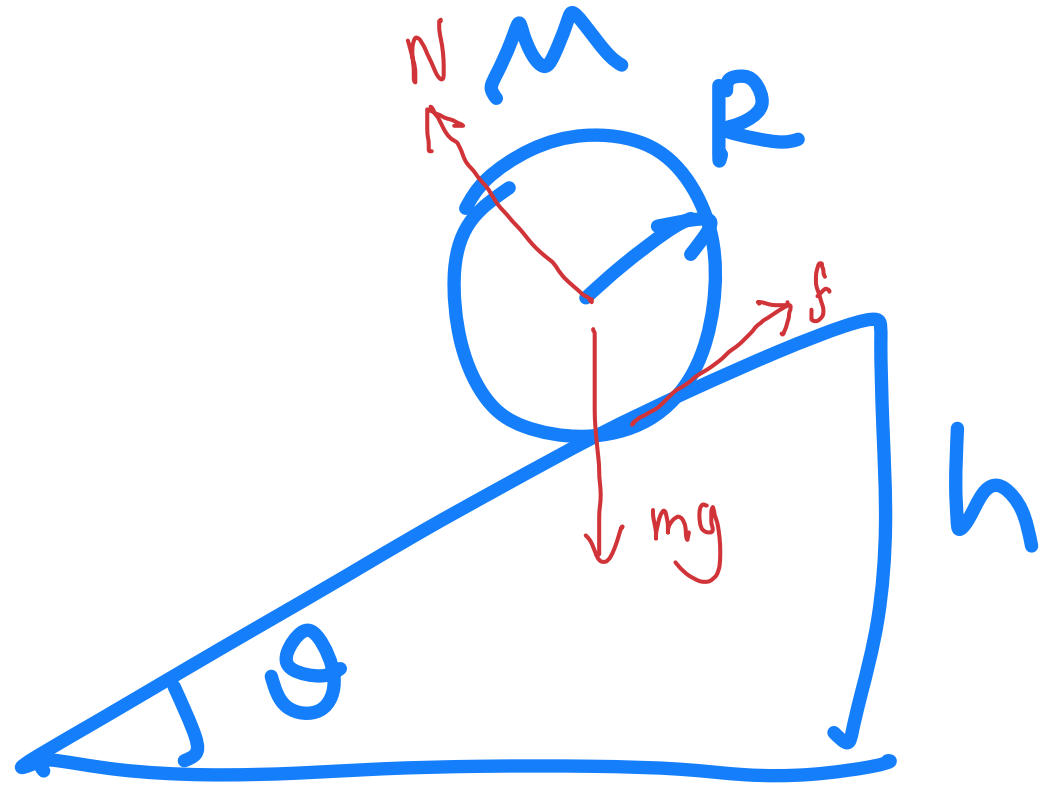
Acceleration of A Rolling Ball



A solid ball of mass M and radius R rolls down a ramp without slipping. The ramp has an angle θ w.r.t. horizontal plane. The surface is not smooth. Find the velocity of center of mass when the solid ball reaches the end of the ramp.

$$\left\{ \begin{array}{l} k = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} m v^2 \\ \omega R = v \\ \Delta k = -\Delta U = mgh \end{array} \right. \quad \begin{array}{l} \text{rotational KE} \quad \text{translational KE} \\ \\ \text{roll w/o slipping} \\ \text{Conservation of ME} \end{array}$$
$$v = \sqrt{\frac{10gh}{7}}$$

Acceleration of A Rolling Ball



A solid ball of mass M and radius R rolls down a ramp without slipping. The ramp has an angle θ w.r.t. horizontal plane. The surface is not smooth. Find the minimum static friction coefficient μ_s between the ramp surface and the ball.

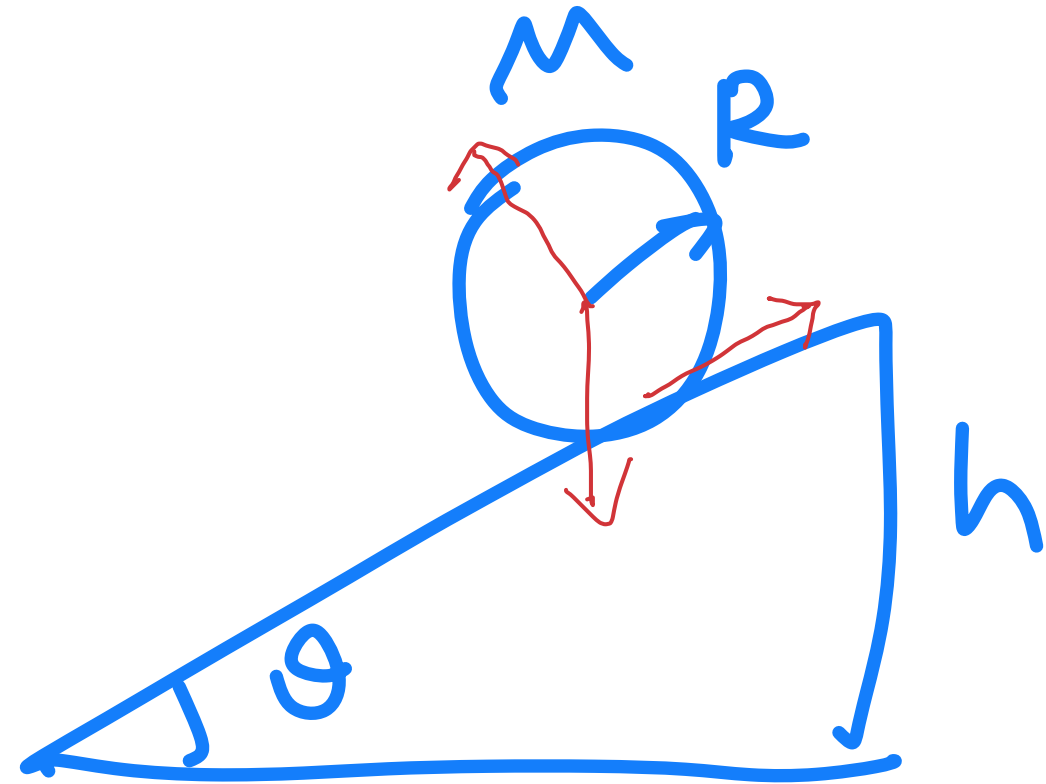
$$\left\{ \begin{array}{l} fR = \frac{2}{5} M R^2 \cdot \alpha \\ Mg \sin \theta - f = Ma \end{array} \right. \quad \left. \begin{array}{l} \text{Newton 2nd law} \\ \text{roll w/o slipping} \end{array} \right\}$$

$$\frac{f}{Mg \sin \theta - f} = \frac{2}{5}$$

$$f = \frac{2}{5} mg \sin \theta \leq \mu_s mg \cos \theta$$

$$\downarrow \mu_s \geq \frac{2}{7} \tan \theta \rightarrow \text{friction}$$

Acceleration of A Rolling Ball



A solid ball of mass M and radius R rolls down a ramp without slipping. The ramp has an angle θ w.r.t. horizontal plane. The surface is not smooth. The static friction coefficient $\mu_s < \frac{2}{7} \tan \theta$, so the ball rolls with slipping. The kinetic friction coefficient is μ_k . Find the velocity of the ball at the end of the ramp.

↳ mechanical energy is not conserved

$$f = \mu_k mg \cos \theta$$

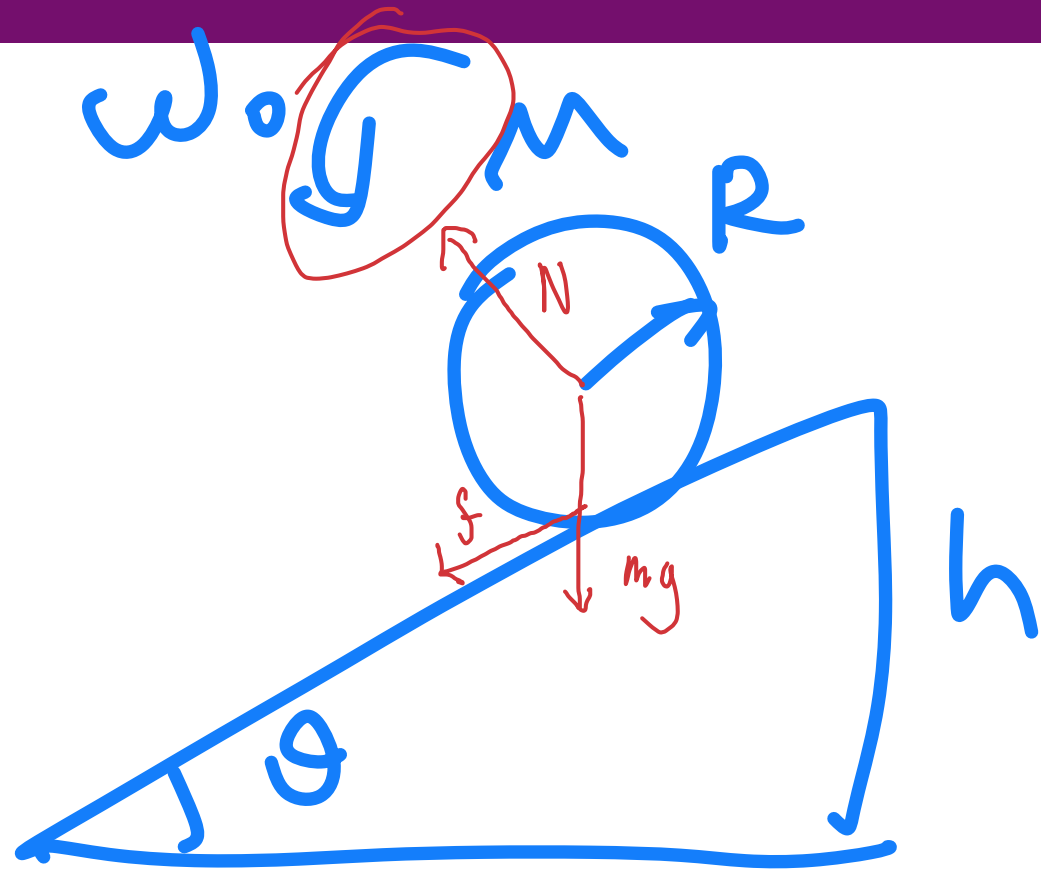
$$\begin{cases} mg \sin \theta - \mu_k mg \cos \theta = ma \\ v^2 = 2as \end{cases}$$

↑
displacement

translation
(Newton 2nd law)
1D motion w/
constant acc.

$$v = \sqrt{2g(\tan \theta - \mu_k)h}$$

Acceleration of A Rolling Ball



A solid ball of mass M and radius R has an initial angular velocity ω counterclockwise, while the center of mass has zero velocity. Then the ball is put on the top of the ramp. The ramp has an angle θ w.r.t. horizontal plane. The surface is not smooth ($\mu_s \geq \frac{2}{7} \tan \theta$). Find the maximum ω , with which the ball can roll without slipping when it reaches the bottom of the ramp.

must hv max value to make the ball not slip

translation

$$mg \sin \theta + \mu_k mg \cos \theta = ma$$

New 2nd law

rotational

$$\mu_k mg \cos \theta \cdot R = \frac{2}{5} m R^2 \alpha$$

already
consider
-ve sign in
counterclockwise
 ω

$$a = g (\sin \theta + \mu_k \cos \theta)$$

$$\omega R = v$$

roll w/o slipping

$$(\omega_0 + \alpha t) R = at$$

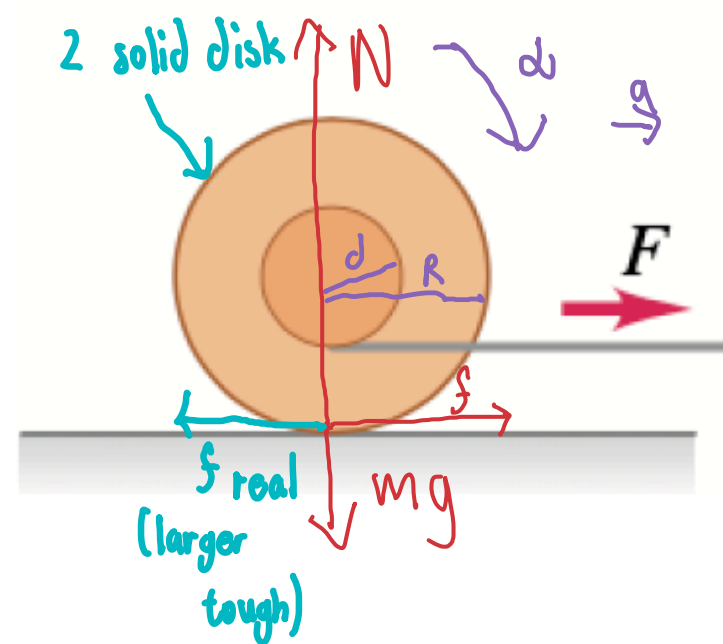
$$\omega_0 = (a - R\alpha)t$$

$$\omega_0 = \frac{\frac{1}{2} at^2}{\int (\sin \theta + \frac{2}{5} \mu_k \cos \theta) \sin \theta} \cdot \sqrt{\frac{2gh}{R}}$$

1D motion

Yo-Yo Revisit

A yo-yo is placed on the surface at rest, and then starts to roll without slipping under force F in each case. Find the direction of rotation in each case. (Hint: find the magic axis.)



$$\begin{cases} F + f = ma \\ -(Fd + fR) = \frac{1}{2} m R^2 \cdot \alpha \\ a = R\alpha \end{cases}$$

$$\frac{F + f}{-(Fd + fR)} = \frac{m a}{\frac{1}{2} m R \cdot a} = \frac{2}{R}$$

$$(F + f)R + 2(Fd + fR) = 0$$

$$f = -\frac{R + 2d}{3R} \cdot F$$

f to the left

Angular Momentum

- $\vec{L} = \vec{r} \times \vec{P}$
 - Unit: $\text{kg} \cdot \text{m}^2/\text{s}$
- $\vec{L} = I\vec{\omega}$, if $\vec{r} \cdot \vec{\omega} = 0$
- $\frac{d\vec{L}}{dt} = \vec{\tau}$ (general, no assumption)

Bullet & Door

A bullet hits and is stuck in a door, as shown below. The door has a length of l and mass m_d . The bullet has a mass m_b and velocity v_b before hitting the door. The bullet is stuck at a position with distance d to the hinge. Find the door's angular speed after the collision.

